Static Optimisation

Many economic models are based on the idea of optimisation. Often choices are constrained by the resources available.

- Consumers choose a consumption bundle $q_1, q_2, \ldots, q_n$ to maximise their utility $u(\mathbf{q})$, subject to the constraint that the total cost should not exceed their budget m:

$$\max_{\mathbf{q}} u(\mathbf{q}) \qquad \text{subject to } p_1 q_1 + p_2 q_2 + \ldots + p_n q_n \leq m$$

- If a firm wants to produce output y, it should choose its inputs of capital and labour in a way that minimises the cost of production:

$$\min_{K,L} rK + wL \qquad \text{subject to } F(K, L) = y$$

General Optimisation Problem

It will be assumed throughout that

- $f(x)$ is a real-valued function of n real variables, $f : \mathbb{R}^n \to \mathbb{R}$.
- S is a subset of $\mathbb{R}^n$, $S \subset \mathbb{R}^n$.
- f is twice continuosly differentiable (C^2) on S; Df is the Jacobian matrix and $D^2 f$ the Hessian:

$$Df = \begin{pmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{pmatrix} \qquad D^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

General Optimisation Problem

The optimisation problem is to maximise (or minimise) f over some subset S:

$$\text{maximise } f(\mathbf{x}), \text{ subject to } \mathbf{x} \in S$$

Maxima

- $\mathbf{x} \in S$ is a global maximum of f on S if $f(\mathbf{x}) \geq f(\mathbf{y})$ for all $\mathbf{y} \in S$.
- $\mathbf{x} \in S$ is a local maximum of f on S if $f(\mathbf{x}) \geq f(\mathbf{y})$ $\forall \mathbf{y}$ in some neighbourhood of $\mathbf{x}$.

These are strict maxima if the inequalities are strict for $\mathbf{x} \neq \mathbf{y}$.

Throughout the lectures we will focus only on maximisation problems. Minimising f is the same as maximising $-f$.

General Optimisation Problem

Recall the Weierstrass Theorem: if S is a compact set, then a continuous f attains a global maximum and global minimum on S.

If S is not compact, the optimisation problem may not have a solution.

When searching for solutions of optimisation problems we need to consider two types of optima:

- an interior optimum: occurs at an interior point of S;
- a constrained optimum: occurs on the boundary of S, where the constraint binds.

If we have an unconstrained optimisation problem, or if the set S is open, we need to look only for interior optima.

Static Optimisation

Topics

Unconstrained Optimisation

When $S = \mathbb{R}^n$ we have an unconstrained problem and there can only be interior maxima.

As with functions of one variable we will need to use first and second order conditions:

First order Condition

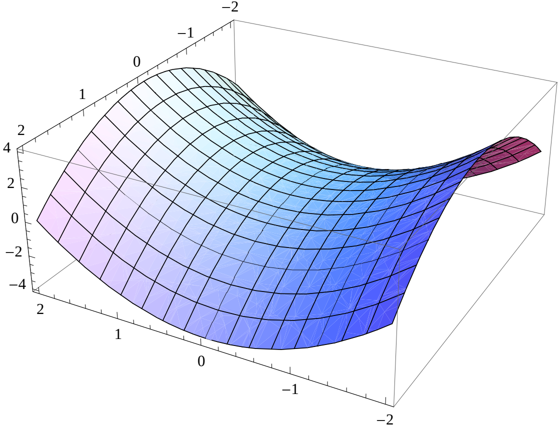
If f has a local maximum or minimum at an interior point $\mathbf{x}^*$ of S, then

$$Df(\mathbf{x}^*) = 0.$$

Unconstrained Optimisation

The possible behaviour of stationary points is more complex in higher dimensions.

Example: Consider $f(x, y) = x^2 - y^2$. $Df(0, 0) = 0$.



Unconstrained Optimisation

Some intuition for the Second Order Conditions comes from the Taylor Expansion for a function of several variables:

$$f(\mathbf{x}^* + \mathbf{v}) = f(\mathbf{x}^*) + Df(\mathbf{x}^*)\mathbf{v} + \frac{1}{2}\mathbf{v}^T D^2 f(\mathbf{x}^*)\mathbf{v} + R(\mathbf{v})$$

At a stationary point $Df(\mathbf{x}^*) = 0$, so if

$$\mathbf{v}^T D^2 f(\mathbf{x}^*)\mathbf{v} < 0 \qquad \text{for all} \quad \mathbf{v} \neq 0$$

then $f(\mathbf{x}) < f(\mathbf{x}^*)$ for all $\mathbf{x}$ close but not equal to $\mathbf{x}^*$, that is $\mathbf{x}^*$ is a local maximum.

When is $\mathbf{v}^T D^2 f(\mathbf{x}^*)\mathbf{v} < 0 \qquad \text{for all} \quad \mathbf{v} \neq 0$?

Recall that the Hessian $D^2 f(\mathbf{x}^*)$ is a square symmetric matrix.

Definiteness

A square symmetric matrix A of order n is

- positive definite if $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq 0 \in \mathbb{R}^n$,
- positive semi-definite if $\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x} \neq 0 \in \mathbb{R}^n$
- negative definite if $\mathbf{x}^T A\mathbf{x} < 0$ for all $\mathbf{x} \neq 0 \in \mathbb{R}^n$,
- negative semi-definite if $\mathbf{x}^T A\mathbf{x} \leq 0$ for all $\mathbf{x} \neq 0 \in \mathbb{R}^n$,
- indefinite otherwise

Unconstrained Optimisation

Suppose $\mathbf{x}^*$ is an interior point of S and $Df(\mathbf{x}^*) = 0$.

Second Order **Sufficient** Conditions

- If $D^2 f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a strict local maximum.
- If $D^2 f(\mathbf{x}^*)$ is positive definite, then $\mathbf{x}^*$ is a strict local minimum.
- If $D^2 f(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is a saddle-point.

Second Order **Necessary** Conditions

- If f has a local max. at an interior point $\mathbf{x}^*$ of S, then $D^2 f(\mathbf{x}^*)$ is negative semi-definite.
- If f has a local min. at an interior point $\mathbf{x}^*$ of S, then $D^2 f(\mathbf{x}^*)$ is positive semi-definite.

Definiteness

How do we check the definiteness of a symmetric matrix A?

The Eigen-Value Test

- A is **positive definite** iff all its eigen-values are strictly positive
- A is **positive semi-definite** iff all its eigen-values are weakly positive
- A is **negative definite** iff all its eigen-values are strictly negative
- A is **negative semi-definite** iff all its eigen-values are weakly negative
- A is indefinite otherwise

Example: $A = \begin{pmatrix} 5 & 4 \\ 4 & 11 \end{pmatrix}$

$$det(A - \lambda I) = \begin{vmatrix} 5-\lambda & 4 \\ 4 & 11-\lambda \end{vmatrix} = (5-\lambda)(11-\lambda) - 16 = (\lambda - 3)(\lambda - 13)$$

Eigen-values: $\lambda_1 = 3 > 0$, $\lambda_2 = 13 > 0$, so A is positive definite.

(Also can use $\det A = \lambda_1 \lambda_2$, $tr(A) = \lambda_1 + \lambda_2$.)

Definiteness

One can test for definiteness without calculating the eigenvalues. Define A_k as the square submatrix of A with only the first k rows and columns retained:

$$A_k = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{k1} & \dots & \dots & a_{kk} \end{pmatrix}$$

The k^{th} leading principal minor is the determinant of A_k.

The Determinant Test

- A is **pd** iff $|A_k| > 0$ for all $k = 1, \dots, n$
- A is **psd** iff $|A_k| \geq 0$ for all $k = 1, \dots, n$ AND all row+col. permutations
- A is **nd** iff $(-1)^k |A_k| > 0$ for all $k = 1, \dots, n$
- A is **nsd** iff $(-1)^k |A_k| \geq 0$ for all $k = 1, \dots, n$ AND all row+col. permutations
- A is indefinite otherwise

Definiteness

Examples:

- The leading minors of $A = \begin{pmatrix} 5 & 4 \\ 4 & 11 \end{pmatrix}$ are 5 and $55 - 16 = 39$, so A is positive-definite.
- The leading minors of $B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ are -1 and 1, so B is negative definite.
- The leading minors of $C = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ are -1 and -1, so C is indefinite.

Steps to Find Optima in Unconstrained Problems

1. Solve the FOCs to find all the candidates for interior optima.
2. Check the SOCs: points where the Hessian is negative (positive) definite are local maxima (minima).
3. Evaluate the function at these points to find a possible global maximum (minimum).

But this procedure does not:

- tell you what happens at a point where the Hessian is negative semi-definite — it may or may not be a local maximum;
- tell you whether the interior local optimum where f is largest or smallest is actually a global optimum — $S = \mathbb{R}^n$, so unbounded (no compact), so the global optimum might not exist.

An Example

Consider the function $f(x, y) = x^3 + y^3 - 3x - 3y + 3xy$.

- Find all the local maxima and minima of f
- Does the global maximum or minimum exist?

1. We start looking for critical points:

   Jacobian:

   $$Df(x, y) = \begin{pmatrix} 3x^2 - 3 + 3y & 3y^2 - 3 + 3x \end{pmatrix}$$

   FOCs:

   $$x^2 - 1 + y = 0$$
   $$y^2 - 1 + x = 0$$

An Example

FOCs:

$$x^2 - 1 + y = 0$$
$$y^2 - 1 + x = 0$$

The first equation implies $y = (1 - x^2)$. Replacing in the second equation we get:

$$\begin{aligned}
(1 - x^2)^2 - 1 + x = 0 \quad &\Longrightarrow \quad (1 - x)^2(1 + x)^2 - (1 - x) = 0 \\
&\Longrightarrow \quad (1 - x)[(1 + x)^2(1 - x) - 1] = 0 \\
&\Longrightarrow \quad (1 - x)x[1 - x - x^2] = 0
\end{aligned}$$

So, either $x = 1$, or $x = 0$, or $x = 1 - x^2 \implies x = \frac{-1 \pm \sqrt{5}}{2}$

An Example

Critical Points:

(a) $x = 1, y = 0$

(b) $x = 0, y = 1$

(c) $x = y = \frac{-1+\sqrt{5}}{2}$

(d) $x = y = \frac{-1-\sqrt{5}}{2}$

2. Check SOCs:

Jacobian: $Df(x, y) = \begin{pmatrix} 3x^2 - 3 + 3y & 3y^2 - 3 + 3x \end{pmatrix}$

Hessian:

$$H = D^2 f(x, y) = \begin{pmatrix} 6x & 3 \\ 3 & 6y \end{pmatrix}$$

An Example

SOCs:

(a) $H = \begin{pmatrix} 6 & 3 \\ 3 & 0 \end{pmatrix}, |H| = -9 < 0 \implies$ indefinite $\implies$ saddle point

(b) $H = \begin{pmatrix} 0 & 3 \\ 3 & 6 \end{pmatrix}, |H| = -9 < 0 \implies$ indefinite $\implies$ saddle point

(c) $H = \begin{pmatrix} 3(-1+\sqrt{5}) & 3 \\ 3 & 3(-1+\sqrt{5}) \end{pmatrix}, |H| = 9(6 - 2\sqrt{5} - 1) = 9(5 - 2\sqrt{5}) > 0,$

$|H_1| = 3(-1+\sqrt{5}) > 0 \implies$ p.d $\implies$ strict local minimum $\quad (f \simeq -2.1)$

(d) $H = \begin{pmatrix} 3(-1-\sqrt{5}) & 3 \\ 3 & 3(-1-\sqrt{5}) \end{pmatrix}, |H| = 9(5 - 2\sqrt{5}) > 0,$

$|H_1| = 3(-1-\sqrt{5}) < 0 \implies$ n.d $\implies$ strict local maximum $\quad (f \simeq 9.1)$

An Example

3. Global Maxima?

   No more interior optima, BUT, we need to check how the function behaves in the limits.Clearly,

$$\lim_{x \to = +\infty} x^3 + y^3 - 3x - 3y + 3xy = +\infty$$

$$\lim_{x \to = -\infty} x^3 + y^3 - 3x - 3y + 3xy = -\infty$$

   So no global solution.

Global Optima: Concavity

In the case of concave functions, stationary points are automatically global maxima.

How do we see whether a multivariate function is concave or convex?

Let f be a C^2 function defined on a convex set S.

- $D^2 f$ is negative semi-definite for all $\mathbf{x} \in S \iff f$ is concave on S.
- $D^2 f$ is negative definite for all $\mathbf{x} \in S \implies f$ is strictly concave on S.
- $D^2 f$ is positive semi-definite for all $\mathbf{x} \in S \iff f$ is convex on S.
- $D^2 f$ is positive definite for all $\mathbf{x} \in S \implies f$ is strictly convex on S.

Note the difference between the local conditions and the global conditions.

Sum Up

- We have seen how to look for optima in unconstrained optimisation problems:
  - Find critical points
  - Check local second order conditions
  - Evaluate the function at those points to find potential global optima
  - Argue whether these are global optima

- Note, that these techniques apply to find any interior optima for any domain S (as long as S has a non-empty interior). However, in those cases the solution (if it exists) might be in the constraint.